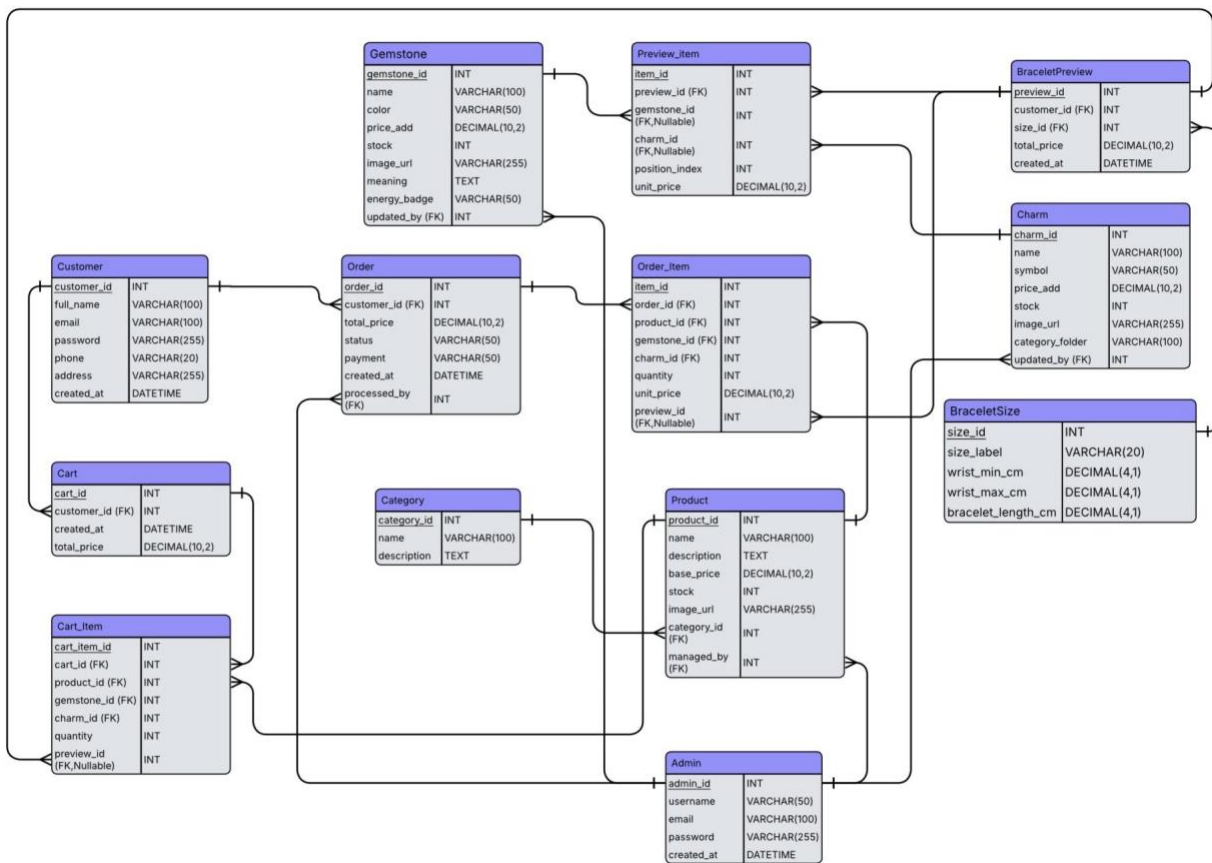


ERD Explanation



1. The Customer Experience (Accounts & Activity)

- **Customer and Cart:** Every customer is assigned one specific shopping cart. This is a one-to-one relationship because the cart is unique to that user's current session.
- **Customer and Order:** A customer can be very loyal and place many different orders over time, but each individual order is linked back to only one specific customer (one-to-many).
- **Customer and BraceletPreview:** This is where the creativity happens. A single customer can design and save as many custom bracelets as they like in their profile (one-to-many).

2. The Customization Logic (The DIY Feature)

- **BraceletSize and BraceletPreview:** We offer several standard sizes. Many different custom designs can use the same size category, such as "Medium" (one-to-many).
- **BraceletPreview and Preview_Item:** This is the heart of your "Design Your Own" feature. One custom design is actually a collection of many individual pieces (stones and charms) lined up together (one-to-many).
- **Gemstone and Preview_Item:** We have a library of gemstones. A single type of stone, like "Blue Zircon," can be picked and placed into thousands of different customer designs (one-to-many).
- **Charm and Preview_Item:** Similar to stones, one specific charm design, like a "Silver Heart," can be used as a component in many different bracelet previews (one-to-many).

3. Shopping and Checkout (Cart & Order Details)

- **Cart and Cart_Item:** A shopping cart acts as a container for many different items that the user is thinking of buying (one-to-many).
- **Order and Order_Item:** Once a purchase is made, the order acts as a permanent record that lists many different items sold in that specific transaction (one-to-many).
- **Product and Cart/Order Items:** A ready-made product from our collection can be added to many different people's carts or appear in many different orders (one-to-many).
- **BraceletPreview and Cart/Order Items:** This is the bridge for your customization. A finished custom design can be added to a cart just like a regular product, and once paid for, it is recorded as a line item in an order (one-to-many).

4. System Management (Administrative Control)

- **Category and Product:** To keep the store organized, we use categories. One category (like "Seasonal Specials") can contain many different products (one-to-many).
- **Admin and Product:** Our team of administrators can manage the catalog. One admin has the authority to add or edit many different products (one-to-many).
- **Admin and Order:** Admins are responsible for fulfillment. One admin can process and update the status of many different customer orders (one-to-many).
- **Admin and Gemstone/Charm:** To keep the DIY tool running, admins manage the inventory. One admin can update the stock levels and prices for many different stones and charms (one-to-many).